



General

FA2026 • 2025-09-24

Physical Security

Tyler

Announcements

- **Fall CTF 2026**
 - Intro hacking competition run by SIGPwny
 - September 27th, 12–6 PM
 - Visit <https://fallctf.com> for more information and to register



ctf.sigpwny.com

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What is Physical Security?

- Deterring threat actors for a physical thing rather than a virtual one
 - Theft/espionage of funds or company secrets, physical harm, etc
 - Very often intersects (IoT/embedded systems, servers)
- Threat models vary widely, what you realistically need is not what a government office needs
 - "Gates, guards, and guns" not the correct answer for everyone



Access Control



- Only allow people you want inside
- Fences, gates, and walls are basic forms of physical access control
- Can get more complicated
 - Keys - suitable for most civilians
 - Keycard entry



Surveillance

- Deters more noticeable forms of entry
 - "Just break a window/door/wall"
 - Lock bypass can be time consuming
- Cameras
 - Vanilla security cameras
 - "Smart" cameras, cloud
- Guards



Can You Spot 5 Barriers to Entry??



Cameras and More Barriers



Security Checks



"Know Your Rights"



- Different states have different laws on owning lockpicks and physical security tools
- Charges can be made worse if lockpicks are found on you
- <https://www.toool.us/lockpicking-laws.php> for more details about about U.S. lockpicking law
- We are not lawyers



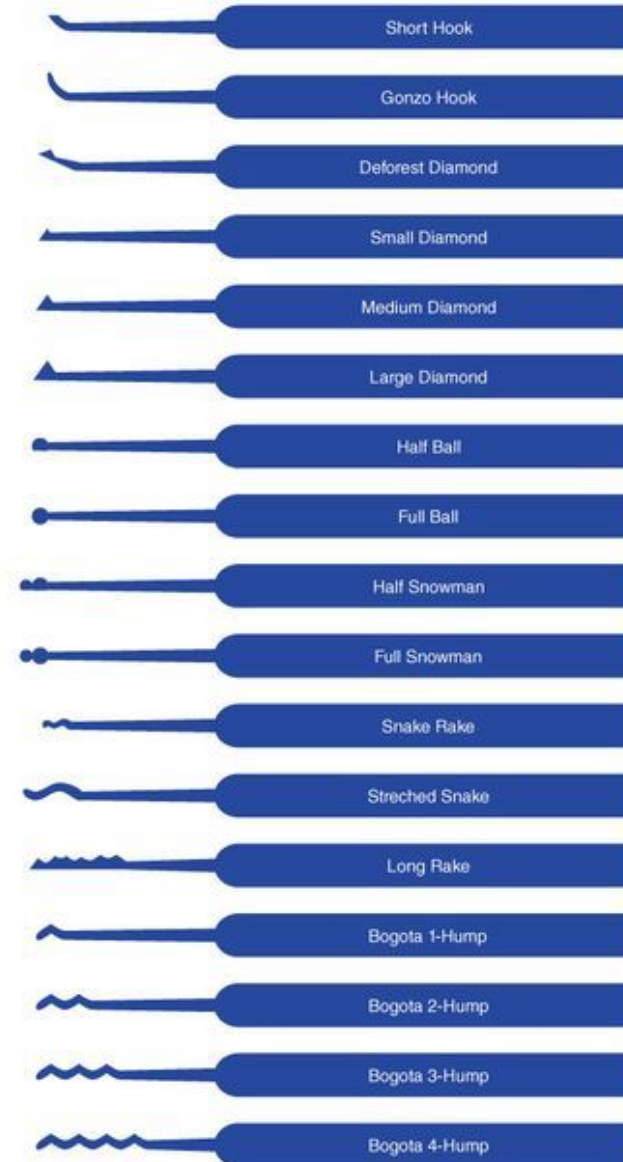
Lockpicking

- Goal: perform the roles of the key without having it
 - Physical object performing the turning action
 - Pins in the spot they need to be to turn the lock
- Manufacturing defects allow you to set pins one at a time without always having the key in place
- Method:
 - Provide tension to turn the keyway using tension
 - Set pins in place using a lockpick

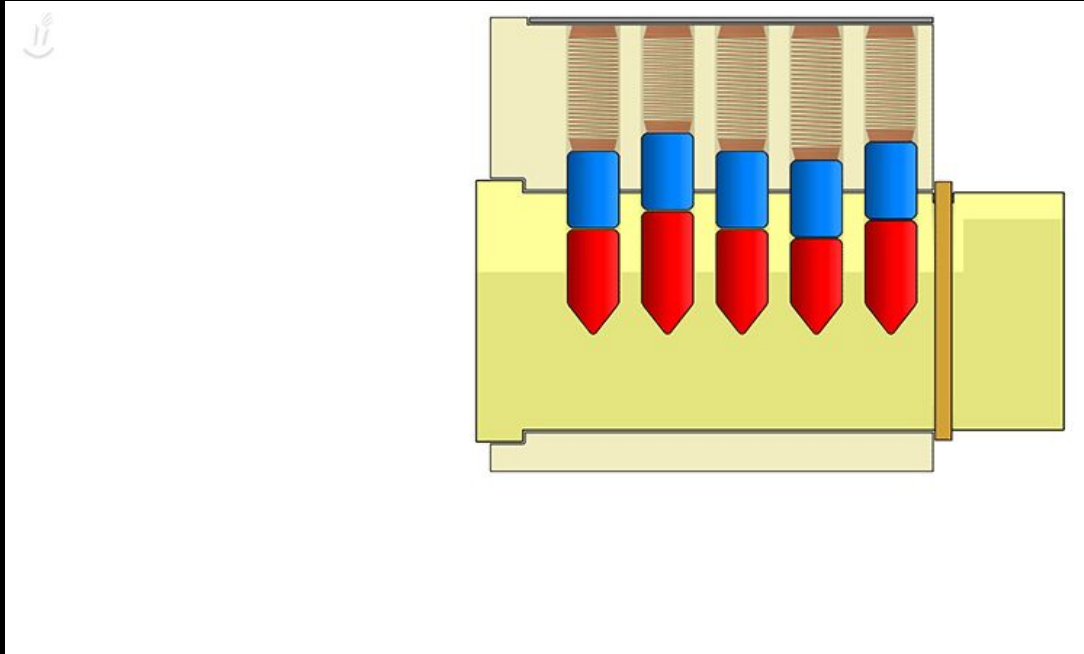


Lockpicks

- Single pin
 - Hooks
 - Diamonds
 - Ball/snowmen
- Rakes
 - Long rakes
 - Snakes
 - Bogata



Single Pin Picking

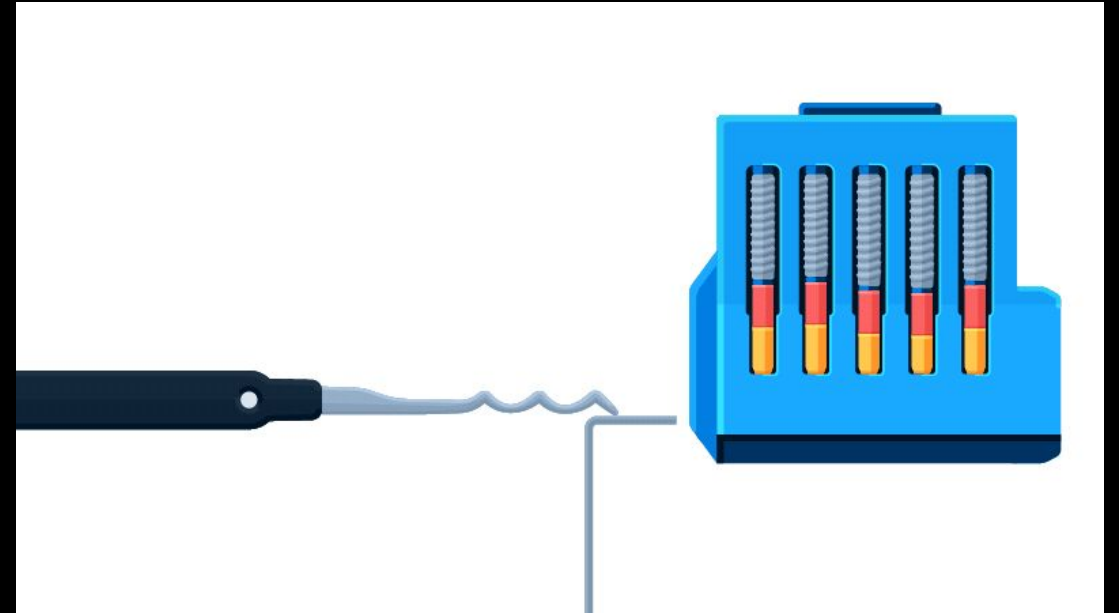


1. Grab a properly sized tensioning tool and a short hook
 - a. The crowbar shaped thing and the hook shaped one
2. Put the tensioning tool either at the top or bottom of the keyway and turn
 - a. Your picks go at the bottom
 - b. Don't push too hard, just hard enough that you'll feel feedback
3. Feel for the pin with the most tension and bring it up until it clicks
4. Continue the process until the lock is picked



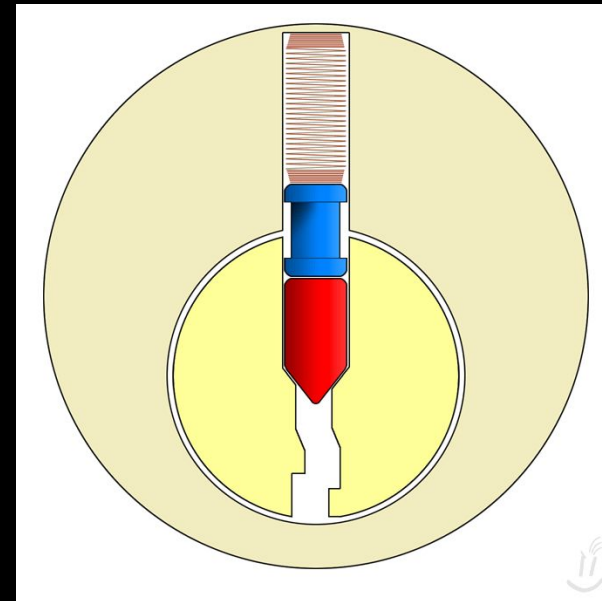
Raking

- The power of probability!
- Gently rock the wavy shaped pick inside the lock until you can turn it open with the tensioning tool
- Very effective with low security locks, becomes less reliable when you face security pins



Security Pins

- Deters low-skilled attacks by making picking harder
- Come in many shapes, all designed to trick you into thinking the lock is partially picked
- Makes lockpicking more fun!



Hardware Security

- Physical security can become digital access
- Only test systems you own or have authorization to access!!!!



Flipper Zero

- Pocket-sized hardware exploration tool
 - NFC/RFID, sub-GHz radio, infrared, and USB
- Can inspect and emulate some supported devices (including keys, fobs, bluetooth devices, etc.)
- Useful for testing access cards, remotes, and IoT systems



What about firewalls?

- Modern networks block inbound connections from the wider-internet by default.
- Most networks will only have specific services explicitly exposed.
- What are different ways we can get access to the network?



Wi-Fi Pineapple

- Portable platform for wireless security testing
- Audit how devices discover and connect to Wi-Fi
 - Look-alike access points to steal credentials (deauth attack)
 - Abuse of automatic reconnection
- Turn physical proximity to initial network access



USB Rubber Ducky

- USB device that identifies itself as a keyboard
- Types a prewritten sequence much faster than a person
- Turns brief physical access into malicious actions on a computer
- Make sure to lock your computer around your friends!



DO NOT GET ARRESTED x3

- We are not responsible for what you do with this information
- Do attempt to use SIGPwny as a scapegoat for your malicious actions
- Take accountability for your actions
- Think before you act!!!!



Next Meetings

2026-10-01 • Next Thursday

- Reverse Engineering II
- Learn how to reverse engineer x86-64 programs

2025-10-04 • Next Sunday

- Cryptography I
- Learn the basics of cryptography, including ciphers, symmetric encryption, and more!



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Meeting content can be found at
sigpwny.com/meetings.

